

Kai Gu

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Education

Bocconi University PhD in Marketing	Milan, Italy 2026 – Present
Bocconi University MSc in Data Science	Milan, Italy 2024 – 2026
Zhejiang University BEcon in Economics	Hangzhou, China 2020 – 2024

Research Interests

Substantive: Economics of Digitization & Platforms, Economics of AI, Creator Economy

Methodological: Causal Inference, Computational Social Sciences, Multimodal Data

Working Papers

Governing AI Content Tools: The Efficiency–Diversity Trade-off in Platform Knowledge Ecosystems

(with Kai Zhu and Lorenzo Lupo)

Knowledge platforms increasingly deploy AI tools to accelerate content production, yet the ecosystem-level consequences of these deployment decisions remain poorly understood. We conduct a large-scale audit of AI-assisted content creation on Wikipedia, one of the earliest major platform deployments of AI content tools, examining how machine translation (MT) relates to information diversity in multilingual knowledge ecosystems. Using over half a million matched article pairs across ten language editions, we document an efficiency–diversity trade-off: relative to independently written (human-created) target-language articles, MT-assisted articles provide more complete coverage for target-language readers but contribute significantly less novel information to the cross-language knowledge ecosystem. MT-assisted articles exhibit lower semantic divergence from their English source articles and lower information breadth across language pairs and topic categories; the missing content is often culturally anchored—local context, regional perspectives, and cultural details that human editors naturally add. Our findings suggest the need for explicit governance frameworks that acknowledge trade-offs, monitor ecosystem-level outcomes, and design AI deployment strategies that preserve valued properties while capturing efficiency gains.

AI Search and Web Traffic

(with Qiaoni Shi and Kai Zhu)

This paper investigates whether AI search restores the referral ecosystem by driving downstream traffic or exacerbates ‘zero-click’ consumption by synthesizing answers within the interface. Leveraging granular clickstream data and a staggered difference-in-differences design, we estimate the causal impact of AI search adoption on users’ broader web consumption and compare the magnitude of direct referral traffic against traditional search engines. We further test for heterogeneity to determine if AI intermediaries disproportionately concentrate traffic toward specific domain types and to analyze the ‘AI divide’ across user demographics. Our findings illuminate the long-term viability of the ad-supported internet, providing insights for policymakers and content publishers navigating the evolving competitive dynamics between AI platforms and the open web ecosystem.

Presentations

AI Search and Web Traffic (with Qiaoni Shi and Kai Zhu)

ISMS Marketing Science Conference, Lisbon, Portugal

June 2026

Governing AI Content Tools: The Efficiency–Diversity Trade-off in Platform Knowledge Ecosystems

(with Kai Zhu and Lorenzo Lupo)

13th Wiki Workshop (Online)

March 2026

References available upon request.